

CYCADS, GINKGOS, AND GNETOPHYTES

Found mainly in warm parts of the world, the cycads, ginkgos, and gnetophytes are three ancient divisions of non-flowering plants with strikingly varied shapes. They range from climbers and low-growing shrubs to plants that are easily mistaken for palms.

DIVISION	CYCADOPHYTA
CLASS	1
ORDER	1
FAMILIES	2
SPECIES	330

DIVISION	GINKGOPHYTA
CLASS	1
ORDER	1
FAMILY	1
SPECIES	1

DIVISION	GNETOPHYTA
CLASS	1
ORDERS	1
FAMILIES	3
SPECIES	70

✓ **BUILT TO LAST**
Like palms, cycads usually have a single crown of compound leaves, surrounding a central growing point, or apical meristem. Their tough leaves are able to withstand strong sunshine and drying winds.

These three groups are traditionally classed, along with conifers, as gymnosperms – a word that means “naked seed”. Unlike flowering plants, gymnosperms form their seeds on exposed surfaces, which are typically specialized scales. Flowering plants produce their seeds in closed chambers, or ovaries.

Scientists have not yet resolved the exact relationship between cycads, ginkgos, and gnetophytes. Nor can they say where they stand in relation to conifers and flowering plants. At the cellular level, some features of gnetophytes suggest they are more closely related to conifers than to cycads and ginkgos, but they also have features in common with flowering plants.

Apart from the nature of their seeds, the three groups have little in common, and hardly ever grow in the same place. Cycads grow mainly in the tropics and subtropics, while the single surviving species of ginkgo comes from China. Gnetophytes are more diverse, including

trees and climbers from the tropics, densely branched shrubs from dry habitats, and the bizarre *Welwitschia*, which grows only in the Namib Desert of southwest Africa.

VARYING FORTUNES
Cycads have a long history, dating back nearly 300 million years. Once an important part of the world’s vegetation, they have gradually lost ground in the face of competition from flowering plants. Today, nearly a quarter of cycad species are endangered, under threat from illegal plant collection and habitat change.

Gnetophytes face fewer problems, and the ginkgo has seen a dramatic improvement in its status. For centuries it was preserved by Buddhist monks, who grew it in temple gardens long after it had vanished in the wild. Introduced to Europe in the 1700s, it proved easy to grow and highly tolerant of polluted air. It is now planted worldwide in parks and city streets.

